



GEETANSH BANSAL



ACADEMIC DETAILS

Year	Degree / Board	Institute	GPA / Marks(%)
---	B.Tech in Mathematics & Computing	Indian Institute of Technology Delhi	9.22
2024	CENTRAL BOARD OF SECONDARY EDUCATION	LALA LAJPAT RAI INTL SCHOOL	97%
2022	CENTRAL BOARD OF SECONDARY EDUCATION	LALA LAJPAT RAI INTL SCHOOL	98.4%

SCHOLASTIC ACHIEVEMENTS

- **National Standard Examination in Physics(NSEP):** Qualified and secured position in National top 1%
- **National Standard Examination in Astronomy(NSEA):** Qualified and secured position in National top 1%
- **National Standard Examination in Junior Science(NSEJS):** Qualified and secured position in National top 1%
- **Indian Olympiad Qualifier in Mathematics(IOQM):** Qualified
- **National Standard Examination in Chemistry(NSEC):** Secured position in State top 1%

QUALIFYING EXAMS

- **Joint Entrance Examination (JEE) Mains Rank:** 276

INTERNSHIPS

- **Bhaasha, Hauz Khas, Delhi** (February, 2025- April, 2025) : *Machine Learning Intern*
 - Developed a pipeline using **LLMs (Gemini APIs)** for **OCR, layout reconstruction, and legal translation** of regional-language documents.
 - Designed system prompts to generate **HTML+JSON structured outputs** from image and OCR data, preserving formatting (tables, alignment, bold/italic tags) with **token-budgeted async chunking**.
 - Built an LLM-based **validation agent** to compare rendered HTML with original images and provide correction feedback via a custom **validationSchema**.
 - Enabled **bidirectional feedback** by back-translating outputs into regional languages to detect semantic drift. Used OpenCV for pre-processing, **LangDetect**, **ReportLab** for PDF generation, and **custom JSON schemas**.

PROJECTS

- **Convex Separator Problem (Prof. Rahul Garg)** (June, 2025 - Present) :
 - Reviewed a paper on how to separate a convex set from another using **k-hyperplanes** and working on how to improve the results
- **Motion Comparison using LLMs & MediaPipe (Prof. Rajesh Khanna)** (April, 2025 - May, 2025) :
 - Developed a motion analysis tool using **MediaPipe** to get 3D joint coordinates from videos across **33 body landmarks**.
 - Used **Gemini 2.0 Flash API** to generate summaries of each subject's motion, **directional change, joint coordination**.
 - Designed a comparative prompt to the LLM to identify **3+ key biomechanical differences**, suggest potential causes (e.g., stiffness, aggression), and assess **physical condition**.
 - Employed **OpenCV for video I/O** and frame handling, json for storing landmark time-series, and **prompt engineering** for effective LLM communication.
- **Computer Vision on CIFAR Dataset(Self Project)** (May, 2025 - June, 2025) :
 - Evaluated image classification methods: **CNNs, Attention mechanisms, ResNets, kNN, Vision Transformers (ViT)**
 - Designed a custom **5-layer CNN architecture** with dropout, batch normalization, and ReLU non-linearities. Tuned **learning rate, optimizer (Adam vs SGD)**, batch size, and regularization. Achieved **~80% accuracy on CIFAR-10**
- **Intelligent Note Maker (Self Project)** (January 2025) :
 - Built a audio-driven classroom companion using **Streamlit** with **CSS UI**, allowing users to record lectures and generate notes. Integrated **PyAudio** for recording, **SpeechRecognition (Google API)** for audio-to-text transcription.
 - Deployed **Gemini 2.0 Flash via Google GenerativeAI** to summarize transcripts into readable, structured lecture notes.
 - Enabled natural language doubt-solving by embedding the generated notes using **all-mpnet-base-v2 (SentenceTransformers)** and querying top relevant chunks from **Qdrant (in-memory vector DB)**.
- **Time Series Forecasting using RNNs & Attention (Self Project)** (March 2025) :
 - Applied **EDA and analysed co-relation matrices** to get necessary features from the dataset of around 90 features
 - Explored results of using **GRU and attention** on the dataset but got best results by **XGBoost**.

TECHNICAL SKILLS

- **Machine Learning(Python):** Pytorch, CNNs, Attention, Vision Transformer, LLMs, RAG, RNNs, XGBoost
- **Programming(Python):** Python, APIs, MediaPipe, OpenCV, Pandas, Streamlit, Qdrant, SentenceTransformers



GEETANSH BANSAL



IIT COURSE

Degree	Institute	CGPA	Dept. Rank
B.Tech in Mathematics & Computing	Indian Institute of Technology Delhi	9.22	17

COURSES DONE

Language & Writing Skill, Intro. To Computer Science, Physics Laboratory, Professional Ethics And Social Responsibility - I, Electromagnetic Waves & Qua.mec., Intro. To Electrical Engg., Intro. To Electrical Engg., Calculus, Linear Algebra & Diffe. Equa., Product Realization Through Manufacturing, Chemistry Laboratory, Engg. Visualization & Comm., Engineering Mechanics, Introduction To Chemistry, Introduction To Engineering, Language & Writing Skill-2, Professional Ethics And Social Responsibility - II

EXTRA CURRICULAR ACTIVITIES

- CAIC Tech GC-24- Overall Runner Up Team Member, Inter Hostel TechGC (2024-25)
- Problem Statement 4- Runner Up Team Member, Inter Hostel TechGC (2024-25)
- CAIC Tech GC Participant, Inter Hostel TechGC (2024-25)